25FEBS202ESE

25 APR 2026

Seat No. : ...25301.....

Student Name : ...[illegible].....

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St. John College of Engineering and Management, Palghar, Maharashtra
Autonomous Institute
End Semester Examinations, April 2026
ASH - SEMESTER II
25FEBS202 - Engineering Chemistry

Time : 1 Hour **Maximum Marks : 30**

Instructions
- All Questions are compulsory
- Assume suitable data wherever required

St. John College of Engineering and Management PALGHAR

		Attempt all Questions	Mrk	Course Outcome	RBT Level
1.	a	Calculate the hardness of given sample of water containing following dissolved salts per litre.CaSO4 = 15.2 mg/l , Mg(HCO3)2 = 2.4 mg/l, MgCl2 = 8.5 mg/l .(Atomic Weight: Ca=40, S=32, O=16, Mg=24, H=1, C=12, Cl=35.5)	5	CO1	L3
		OR			
	b	i. Write a Reaction involved in the process of ion exchange method.	3	CO1	L2
		ii. Give the chart of different types of impurities present in the water.	2	CO1	L2
2.	a	Find the atom economy of the following reactions- C6H5CHO + CH3CHO → C6H5CH=CHCHO + H2O (Atomic Weight: , O=16, , H=1, C=12,)	5	CO2	L3
		OR			
	b	i. Explain greener route of production of Carbaryl.	3	CO2	L2
		ii. Explain Avoid chemical derivatives principle of green chemistry.	2	CO2	L1
3.	a	What is electrochemical corrosion? With the help of diagram and electrode reaction explain mechanism of electrochemical corrosion of iron by oxygen absorption in neutral medium.	5	CO3	L2
		OR			
	b	i. Define Corrosion and Classify the various metal oxide films formed during oxidation corrosion.	3	CO3	L3
		ii. Explain how position of metal in electrochemical series affects rate of corrosion	2	CO3	L2
4.	a	A sample of coal contains: C =65%, H= 15%, O = 10%, S= 4%, N=2% and ash=4%. Calculate the minimum amount by weight and volume of air required for complete combustion of 2kg of fuel.(Mol wt of air = 28.949g)	5	CO4	L3
		OR			
	b	i. Give Any 3 Characterstics of good Fuel.	3	CO4	L2
		ii. In a experiment using Bomb calorimeter 1.5g of coal sample gave 0.28g of BaSO4. Calculate percentage Sulphur in the sample.	2	CO4	L1
5.	a	Explain the purpose of making alloys.	5	CO5	L2
		OR			
	b	i. Define conducting polymer and Give the chart of classification of conductiong Polymer.	3	CO5	L2
		ii. Distinguish between Brass and Bronze	2	CO5	L2
6.	a	With the help of a neat labelled diagram, explain function of Burner and Filter/Monochromator and Detector.	5	CO6	L2
		OR			
	b	i. Explain any three characteristics of Electromagnetic radiation.	3	CO6	L2
		ii. Define 1) Absorption Spectrum 2) Emission Spectrum	2	CO6	L1

* All the Best *